October 13, 2021

Dear Mr. True,

The U.S. Food and Drug Administration's (FDA or Agency) Coronavirus Disease 2019 (COVID-19) testing guidance describes a policy for laboratories and commercial manufacturers to help accelerate the availability and use of tests they develop in order to achieve more rapid and widespread testing capacity in the United States. With respect to commercial manufacturers of tests intended to detect SARS-CoV-2 RNA, the policy states in part that FDA does not intend to object to their development and distribution where the test has been validated and while the manufacturer is preparing its emergency use authorization (EUA) request, the manufacturer gives notification to FDA, certain labeling information is included in the test reports and instructions for use, and an emergency use authorization (EUA) request is submitted within 15 business days of the notification to FDA. This guidance was most recently updated on May 11, 2020. As outlined in the policy referenced above, you have previously represented to the Agency that your molecular diagnostic test is validated, and notified the Agency that you intend to distribute, a molecular diagnostic test to identify SARS-CoV-2 RNA and submitted a request for an EUA, EUA210582, for that test.

You have not provided complete information to demonstrate that the studies performed are adequate to validate your test and support the claimed performance characteristics for your device. Specifically:

1. **Clinical Evaluation Supporting the Request for Authorization of the FloodLAMP QuickColor COVID-19 Test:**

   As per the FDA's Molecular Diagnostic Template, validation of a new test should include testing nasopharyngeal (NP) swab specimens from at least 30 positive and 30 negative individuals from the intended use population to support an NP swab or general upper respiratory claim for individuals suspected of COVID-19 by their healthcare providers. To support a general upper respiratory tract (URT) specimen claim it is recommended that the clinical study consists of at least 50% NP swabs and 50% nasal swabs. In your EUA submission you provided results from a clinical validation study that included 40 comparator positive, and 40 comparator negative nasal swab samples. However, the information you provided does not support an NP swab or other common URT specimens claim for the FloodLAMP QuickColor COVID-19 Test for the following reasons:

   a. **Intended Use Population**
   Your test is intended for individuals suspected of COVID-19 by their healthcare providers ("suspected" individuals). However, you have not provided a description of your study population. Therefore, it is unclear whether your clinical study was performed with samples from the intended use population. Consequently, we cannot determine whether your clinical study is supportive of your intended use.

b. Sample Types:
Per your Intended Use, you are claiming upper respiratory specimens including oropharyngeal (OP), NP, anterior nasal (AN), and mid-turbinate nasal swabs. As per the FDA's Molecular Diagnostic Template, studies supporting a NP or general upper respiratory specimen claim should be performed with NP swabs as FDA considers this to be the most challenging upper respiratory matrix. However, your clinical study was performed with the AN swab matrix. As such your claim for NP and OP swabs is not currently supported by your clinical evaluation study.

c. Low performance:
You achieved a positive percent agreement (PPA) and a negative percent agreement (NPA) between your test and the comparator tested nasal swab specimens of 90% (36/40) and 100% (40/40), respectively. However, as per the FDA's Molecular Diagnostic Template, FDA believes a minimum of 95% PPA and NPA is an acceptable clinical performance for a test with an intended use as you proposed for your device. A PPA of 90% is below the performance FDA believes is acceptable. The data obtained in your study suggest clinical performance issues with your device, which may put patients at unreasonable risk of harm due to inaccurate results. False negative test results could have a number of consequences, including leading to additional unnecessary diagnostic evaluations, and could negatively impact the effectiveness of infection control activities by incorrectly assessing the infection status of individuals.

d. Low positive samples:
According to the FDA's Molecular Diagnostic Template, specimens representing a wide range of viral load including low positive specimens should be included in the clinical evaluation. It is recommended that approximately 25% of the SARS-CoV-2 comparator positive specimens should have Ct values close to the mean Ct value of the comparator at its confirmed LoD concentration (i.e., within 3 Cts of the mean Ct value at the confirmed LoD) to be considered as challenging samples with low viral load. However, your clinical data set indicated that no low positive samples per the comparator test were tested in your clinical evaluation study. Without data generated from testing a sufficient number of low positive samples with low viral load, we cannot appropriately assess the performance of the FloodLAMP QuickColor COVID-19 Test.

To address the low performance issue, you should perform a root cause analysis and take the steps necessary to resolve the low PPA for your test. In your response, you should provide evidence that the steps taken addressed the issue and improved the clinical performance. If you identify any new risks in your root cause investigation, you should make sure to provide information to show that those risks have been mitigated. Please note that, should any mitigations require you to alter your device design or testing procedure in any way, additional clinical and analytical validation studies with your final device design may be needed.

Once the sensitivity of your test is improved, to support an NP swab or general URT specimens claim for individuals suspected of COVID-19 by their healthcare providers

using the FloodLAMP QuickColor COVID-19 Test, please provide clinical validation data from an adequate number of positive and negative samples collected from your intended use population that demonstrate acceptable performance when compared to the results obtained with the comparator test. Please ensure that you provide a clear description of how samples in your clinical study were sourced, including the population characteristics of your study population and the sample numbers for each sample type. For a general URT sample claim as proposed, at least 50% of samples in your clinical study should be NP swab samples. Alternatively, you may adjust your specimen claim to claim AN swabs only if these are easier to source. Please also ensure that your clinical validation data set includes an adequate number of challenging/low viral load specimens (i.e., approximately 25% of the SARS-CoV-2 positive swab specimens, defined as those specimens having Ct values within 3 cycles of the mean Ct value at the comparator's LoD). Without the additional clinical validation data requested, FDA is unable to determine that the known and potential benefits of your FloodLAMP QuickColor COVID-19 Test outweigh the known and potential risks.

2. **Inclusivity Study and Variant Analysis**

You performed an *in silico* inclusivity analysis for your test in February 27, 2021. This *in silico* analysis is no longer representative of the circulating strains and variants at this time. Without an updated analysis for your test, we are unable to assess the performance of your device with the currently circulating SARS-CoV-2 strains and variants. Consequently, we are unable to determine that the benefits of your device outweigh the risks. Please provide an updated inclusivity analysis to support the current number and types of variants and provide the information in the table format below. Furthermore, please provide a specific risk assessment related to the performance of your device with all currently circulating variants of interest and variants of concern.

Please provide your updated *in silico* analysis results in the following formats:

Primer	Variant Name	Sequence	Variant Sequence (Mutation Represented by X)	Seq* start	Seq* send	count (Past 30 Days) 30439 U.S. sequences	Frequency (Percent) (Past 30 Days) 30439 U.S. sequences	count (Past 14 Days) 9452 U.S. sequences	Frequency (Percent) (Past 14 Days) 9452 U.S. sequences

*position in the genomic sequence

	Target gene Fwd Primer 1	Target gene REV Primer	Target gene Probe
Total primer length (nt)			
Total strains			
100% match			
1 mismatch			
2 mismatches			
3 mismatches			
>3 mismatches			

3. Cross Reactivity / Microbial Interference

You have provided a cross reactivity study that consists of wet testing as well as a comprehensive *in silico* analysis. However, some organisms with >80% of homology revealed in your *in silico* analysis in Table 13 of your submission were not tested in your wet testing (e.g. *Candida albicans*, *Pseudomonas aeruginosa*, *Legionella pneumophila*, and *Streptococcus pneumoniae*). Therefore, additional information on potential cross-reactivity is needed to support your EUA. In your response to this deficiency, please provide your *in silico* cross-reactivity data in tabular form identifying the pathogen, strain, accession # (e.g., you may pick one that represents the complete genome of the organism), and individual % homology of F/R primers and probes for all targets across organisms evaluated (<u>not only organisms with ≥ 50% of homology</u>). For all organisms that produce homologies to any of the primers and probes of above 80%, microbial interference testing should be provided as described below or provide a rationale for why such testing can be omitted.

Microbial interference should be tested in the absence and presence of SARS-CoV-2 at no more than 3x LoD and the presence of the potentially interfering organism at high concentration (i.e., viruses ≥ 10^5 copies, PFU, or $TCID_{50}$/mL, bacteria and fungi ≥ 10^6 CFU/mL).

Table X. *In silico* cross-reactivity testing with the % homology of the FloodLAMP QuickColor COVID-19 Test primers and probes

	GenBank Accession #	Target 1 Primer -F	Target 1 Primer-R	Target 1 Probe
Human coronavirus 229E				
Human coronavirus OC43				
Human coronavirus HKU1				
Human coronavirus NL63				
SARS-coronavirus				
MERS-coronavirus				
Adenovirus (e.g. C1 Ad. 71)				
Human Metapneumovirus (hMPV)				
Parainfluenza virus 1-4				
Influenza A & B				
Enterovirus (e.g. EV68)				
Respiratory syncytial virus				
Rhinovirus				
Chlamydia pneumoniae				
Haemophilus influenzae				

Legionella pneumophila				
Mycobacterium tuberculosis				
Streptococcus pneumoniae				
Streptococcus pyogenes				
Bordetella pertussis				
Mycoplasma pneumoniae				
Pneumocystis jirovecii (PJP)				
Candida albicans				
Pseudomonas aeruginosa				
Staphylococcus epidermis				
Streptococcus salivarius				

4. **Endogenous Interfering Substances**:

You have provided an endogenous interfering substances study and stated that the potential interfering substances listed in Table 15 were tested at the indicated concentrations in your interfering substances study. However, the potential interfering substances tested in your study are insufficient to support the interference by commonly used cold remedies and products used in the URT. For a general URT specimen claim, FDA recommends evaluating the additional potentially interfering substances listed below for interference with your test because they can be expected to be commonly found in the URT. In your response, please submit interfering substance study data of the additional substances below to support that the tested substances do not interfere with your test. If new interference is observed, in your response, please provide information to show that any risk to patients has been mitigated. Please refer to the table below for our recommendations on the list of endogenous interfering substances that should be assessed to support your intended use. Each condition should be tested with 3 replicates.

Potential Interfering Substances	**Active Ingredients**	**Concentration**	**Negative Samples**	**Positive Samples**
			Detected	**Detected**
Respiratory Samples				
Nasal allergy spray	Triamcinolone acetonide			
Nasal congestion spray	Oxymetazoline HCl	15% v/v		
Flonase	Fluticasone propionate			
Saline nasal spray	NaCL, Phenylcarbinol, Nemalkonium Chloride			

NeilMed Nasogel		1.25%		
Cepacol Lozenges (benzocaine/menthol)	Benzocaine,	3 mg/mL		
Cough drops	Dextromethorphan HBr			
Robitussin				
Chloroseptic Sore Throat spray	Phenol, Glycerin	5% v/v		
Emergen-C	Zinc, Magnesium, Riboflavin, Vitamin C			
Zinc (for pyrophosphate based LAMP assays)	Zinc			
Nyquil	Acetaminophen, Doxylamine succinate, Dextromethorphan HBr			
Mucin: bovine submaxillary gland, type I-S		2.5 mg/ml		
Human Genomic DNA		10 ng/µl		
Vaseline	Petroleum Jelly			
Nicotine	Nicotine	0.03 mg/ml		
Tobacco	Nicotine, Tar, Carbon Monoxide, Formaldehyde, Ammonia, Hydrogen Cyanide, Arsenic, and DDT			
Alcohol	Ethanol	5%		
White blood cells/Leukocytes		1 to 5x10^6 cells/mL		
Whole Blood		2.5%		

5. **External Controls**:

FDA recommends that external positive control material is provided in a concentration that is adequately representative of a real clinical sample and that is challenging enough for the test system to detect potential reagent deterioration. Therefore, it is recommended that the positive control mimics a low positive sample (i.e., ≤ 5x LoD). However, the provided information in your EUA template indicates that you are providing the external positive

controls with a concentration of 100,000 copies/mL equivalent to 8x LoD. Your control does not meet the recommended concentration and is therefore not an adequate control for the performance of your test with low positive samples. Please provide the validation data for the control material for FDA review.

6. **RUO Reagents**:

   In your manufacturing information you indicate that your test ought to be used with recommended Research Use Only (RUO) reagents manufactured by a 3rd party. E.g., LGC Biosearch Technologies/Eurofins Genomics/Integrated DNA Technologies/Sigma LAMP primers, New England Biolabs Colorimetric LAMP Master Mix). As you are aware, RUO products are not subject to 21 CFR Part 820. Because of this, RUO reagents may vary across different manufacturers and lots in ways that may affect the performance of the test. Since continued quality and adequate performance of your test is not assured in this case, you may not recommend as components of an authorized test materials/components that are labeled RUO. Therefore, you have the following options:

   a) Identify and recommend IVD labeled materials from a third-party manufacturer, or

   b) Qualify specific lots of the recommended RUO labeled third party manufactured materials for adequate performance with your test and post the lot numbers e.g., on your product website, or

   c) Bring the materials under your own quality system by supplying them as part of the kit. In this case the control material you purchase is considered a raw material that becomes a component of your kit for which quality and performance you are responsible.

Without the information outlined above, we are not able to determine that your test is adequately validated for its intended use. As a result, the performance of your device is unknown which may put patients at unreasonable risk of harm due to inaccurate results.

By October 15, 2021, 12 pm (eastern), please reply to this email with information sufficient to demonstrate that the device is adequately validated for its proposed intended use, including an explanation of why the new information is adequate to overcome our concerns about the health risks posed by your device's unknown clinical and analytical performance as reported in your original submission.

If you do not yet have the information requested above, it may take significant time to generate it. You have the option to voluntarily withdraw your EUA request while you collect appropriate validation data and resubmit your request once you have adequately validated your device. If you choose to withdraw, please notify us of that in your response.

If you do not respond by October 15, 2021, 12 pm (eastern), or if the information you provide does not demonstrate that your device is adequately validated for its intended use, we may take steps as

described in the guidance referenced above, including removing the test from the website listing of notifications, and likely would not be able to determine that the criteria for emergency use authorization have been met. If such steps are taken and we make that determination, we would expect you to suspend distribution of your test, and we may request that you take additional actions to protect the public health as appropriate.